

LAVA: A Lightweight Benchmarking Framework for Robust and Real-Time Deepfake Voice Detection

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Abstract. Deepfake-voice detectors are often compared across incompatible partitions, score conventions, preprocessing, and timing boundaries. LAVA is a registry-based framework that evaluates heterogeneous detectors through common integrity, score, robustness, and runtime contracts while preserving native architectures. We evaluate four LAVA-trained Mel-sequence models MobileNetV3Small-LSTM, ShuffleNetV2-LSTM, MnasNet-A1-LSTM, and EfficientNet-B0-LSTM and externally pretrained RawNet2 and AASIST references. SHA-256 grouping quarantined 30 byte-identical cross-label files and produced a checksum-group-disjoint manifest; clean evaluation used all 2,737 test recordings. ShuffleNetV2 achieved the best clean result (F1 0.9824, AUC 0.9929, EER 1.46%), whereas MobileNetV3 had the lowest single-thread CPU end-to-end latency (43.8 ms); ShuffleNet required 62.5 ms (RTF 0.0208). A fixed stratified 100-recording diagnostic evaluated seeded noise, four codec round trips, and simulated replay. Noise was the principal weakness; ShuffleNet’s mean F1 degradation across nine conditions was 0.1623. The exploratory Pareto set contained MobileNetV3, ShuffleNetV2, and AASIST. Heterogeneous training provenance and input contracts preclude an architecture-only ranking, and the evidence does not establish full-test robustness, physical replay, unseen-domain, edge-device, streaming, or multi-seed performance.

Keywords: Deepfake voice detection, audio anti-spoofing, lightweight deep learning, robustness, real-time factor, Pareto analysis.

1 Introduction

Synthetic and converted speech threaten speaker verification and human decision making. ASVspoof established shared logical- and physical-access protocols [1], while WaveFake broadened public generated-audio resources [4]. Yet reported performance remains difficult to compare when detectors use opposite class orders, incompatible durations, different preprocessing, or inconsistent runtime boundaries. Byte-identical files crossing data splits create a separate leakage risk.

LAVA is concerned with the evaluation interface rather than making each detector use a single architecture. It retains the original computation of Mel-sequence, raw waveform, recurrent and graph attention models while unifying artifact verification, P(FAKE), integrity test, saved output, stress inputs, timing and aggregations. Current data includes four locally trained lightweight artifacts and two pretrained artifacts; clean evaluation is full testing, while robustness is diagnostic [5], [6]. We ask: Three main research questions (RQs) are answered in this paper:

- RQ1: How competitive are lightweight temporal CNNs against the two references under one clean protocol?
- RQ2: How does performance change under the executed noise, codec, and simulated-replay conditions?
- RQ3: which detectors are non-dominated when EER, robustness degradation, and end-to-end RTF are considered jointly?

Contributions are: (i)

Table 1: Effective placement of sample work. P indicates partial coverage

Work	Robust	Efficiency	Cross-domain	Main relation/limitation
ASVspoof [2], [3]	P	-	P	Standard protocols; no unifying LAVA runtime
RawNet2/AASIST [7], [8]	P	P	-	References external to this study
SSL anti-spoofing [12], [13]	P	-	P	Family of future larger detectors
Cross-domain studies [14], [15]	Yes	P	Yes	Encourage limited scope
LAVA	Yes	Yes	-	Six detectors; unified contract; no invisible set

2 Related Work

Reproducible anti-spoofing methodologies were set forth by ASVspoof and t-DCF [1], [3].

While WaveFake and survey works illustrate additional synthesis methods and detection frameworks [4], [5].

RawNet2 models waveforms directly, whereas AASIST integrates spectral and temporal graph attention [7], [8].

LCNN and SincNet illustrate complementary time-frequency and learned-filter approaches [10], [11].

MobileNet, ShuffleNet, EfficientNet, and MnasNet pursue efficiency through distinct search and scaling principles [19], [21]. Their published vision or phone measurements are not evidence of LAVA audio runtime. Table I positions the current contribution: robustness, resource measurement, integrity, and heterogeneous score semantics are evaluated jointly, while training provenance remains explicit.

3 METHODOLOGY

3.1 LAVA Framework and Experimental Protocol

LAVA separates canonical data and integrity (D), registered detectors (M), executed conditions (C), measurements (E), and analyses (A). The registry declares framework, input, duration, artifacts, threshold, and provenance; adapters retain native computation but expose loading, bounded scores, parameter count, and size. The clean and stress runners enter sample-by-sample scores prior to doing the statistics and graphics, avoiding planned experiments from showing up as results. Figure 1 illustrates the six-detector border.

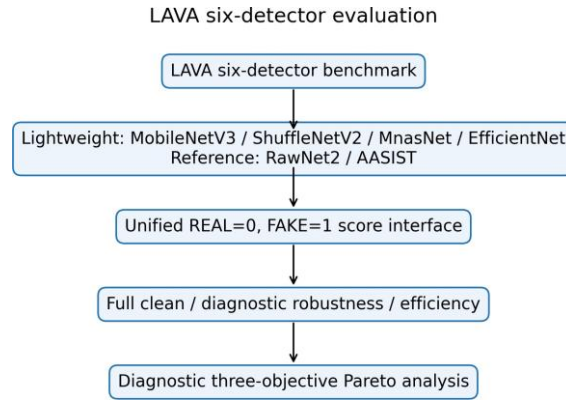


Fig. 1. Six native detectors connected to the unified LAVA evaluation boundary

Checksum-aware canonical dataset construction

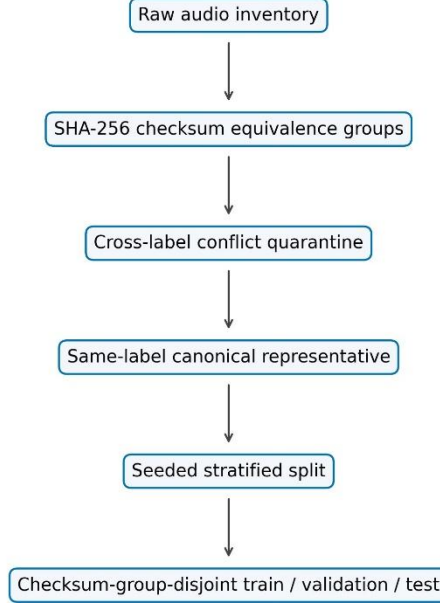


Fig. 2. SHA-256 conflict quarantine, canonicalization, and disjoint splitting.

3.2 Dataset Integrity and Canonical Split

The internal binary collection has no speaker, source, generator, parent-recording, or dataset identifiers; LAVA therefore claims neither speaker, source, generator, nor cross-dataset independence. It scanned 18,722 files (10,550 REAL, 8,172 FAKE), found 435 duplicate groups and 476 redundant files, and quarantined all 30 files in 14 cross-label checksum groups. Within a same-label group, the lexicographically first path is canonical. The retained 18,232 recordings comprise 10,493 REAL and 7,739 FAKE, split with seed 42 into 12,762/2,733/2,737 train/validation/test samples. The supported claim is checksum-group-disjoint only. For $h(x) = \text{SHA256}(x)$ and split group sets G_s ,

$$x_i \sim x_j \Leftrightarrow h(x_i) = h(x_j), \quad G_{tr} \cap G_{va} = G_{tr} \cap G_{te} = G_{va} \cap G_{te} = \emptyset.$$

The manifest hash is:

8b55591d58d3658b8cafe0e77b6ebdedbaa67be2e339730a7276fe9b10958df9.

Table 2: Canonical manifest and split inventory

Inventory	Count	Split	Count
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Scanned / retained	18,722 / 18,232	Train	12,762
Retained REAL / FAKE	10,493 / 7,739	Validation	2,733
Duplicate groups / redundant	435 / 476	Test	2,737
Cross-label groups / files	14 / 30	Independence	checksum groups

3.3 Standardized Audio Preprocessing

Lightweight inputs are decoded, channel-averaged, polyphase-resampled to 22,050 Hz, and padded/truncated to 3.0 s. Six chronological 0.5-s segments use a Hann STFT ($N = 2048$, hop 512), 128 HTK-style Mel filters spanning 20–8,000 Hz, relative dB mapping clipped to 80 dB, $[0, 255]$ scaling, bilinear 224×224 resizing, and RGB replication [24]. For segment t

$$X_t(m, k) = \sum_{n=0}^{N-1} x_t[n + mH]w[n]e^{-j2\pi kn/N}, \quad S_t^{mel}(m, r) = \sum_k |X_t(m, k)|^2 H_r(k)$$

3.4 Evaluated Detector Architectures

For the lightweight family, shared segment encoder f_θ produces six embeddings, an LSTM with 128 units aggregates them, and a 64-unit ReLU layer plus dropout 0.4 feeds a sigmoid:

$$p_{fake} = \sigma \left(W_2 \text{Dropout}_{.4} \left(\text{ReLU} \left(W_1 \text{LSTM} \left(f_\theta(X_{1:6}) \right) + b_1 \right) \right) + b_2 \right). \quad (3)$$

3.5 Training, Artifact Provenance, and Score Semantic

MobileNet and EfficientNet use ImageNet initialization; EfficientNet’s deployment is explicitly the best warm-up checkpoint, not a completed fine-tuning lifecycle. MnasNet and ShuffleNet were trained jointly with their recurrent heads from scratch; all 49 and 56 BatchNormalization layers, respectively, were trainable from epoch 1. RawNet2 and AASIST were not trained by:

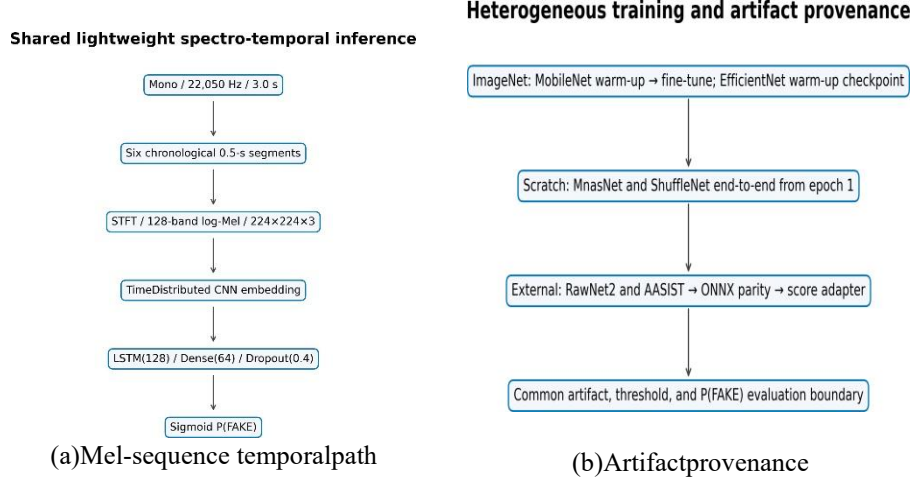


Fig. 3. Architecture and provenance: the evaluation contract is shared, training is not.

Table 3: Detector specification, provenance, and deployed threshold.

Detector	Input	Temporal core	Params	Initialization/artifact provenance	τ
MobileNetV3-LSTM [18], [19]	Mel, 3.0 s	LSTM(128)	1,308,401	ImageNet; warm-up + partial fine-tune	.82
ShuffleNetV2-LSTM [20], [21]	Mel, 3.0 s	LSTM(128)	1,868,441	scratch; full end-to-end, best epoch 16/28	.12
MnasNet-A1-LSTM [23]	Mel, 3.0 s	LSTM(128)	3,369,255	scratch; full end-to-end, best epoch 27/39	.90
EfficientNet-B0-LSTM [22]	Mel, 3.0 s	LSTM(128)	4,779,300	ImageNet; best warm-up epoch 47	.90
RawNet2 [7]	waveform, 4.04 s	GRU	17,621,410	external reference; ONNX export	.50
AASIST [8], [9]	waveform, 4.04 s	graph readout	297,866	external reference; ONNX export	.50

LAVA: strict PyTorch loading followed by ONNX export gave maximum native/export score differences of 4.11×10^{-6} and 5.96×10^{-8} on six real tensors. This verifies conversion, not identical training or publication-pipeline equivalence.

LAVA fixes REAL=0 and FAKE=1; every adapter returns $p = P(\text{FAKE})$ and

$$\hat{y} = \mathbf{1}[p \geq \tau], \quad \tau^* = \arg \max_{\tau} F 1_{val}(\tau).$$

The four low-weight thresholds in Table III are validation-only. No calibration on external references. Defaults on those are 0.5, uncalibrated. Labels do not influence threshold choice or weighting. Score values are not claims of calibrated real-world probability.

3.6 Clean and Robustness Evaluation

The clean analysis makes use of all 2,737 testing samples (1,575 REAL, 1,162 FAKE). Accuracy, classwise and macro F1, ROC-AUC from raw scores, and linearly interpolated EER are reported. With FAKE positive,

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F1 = \frac{2PR}{P + R}; \quad FAR = \frac{FP}{FP + TN},$$

$$FRR = \frac{FN}{FN + TP}, \quad EER \simeq \frac{FAR + FRR}{2}.$$

Robustness uses a preselected stratified subset of 100 test recordings (58 REAL, 42 FAKE; seed 42), and each baseline uses those same IDs. Four seeded AWGN levels (20, 10, 5, 0 dB) use sample-identity-derived seeds. FFmpeg round trips cover MP3 128/64 kb/s, Opus 64 kb/s, and AAC 96 kb/s. Simulated replay applies taps at 0/17/43/89 ms with gains 1/.45/.25/.12, normalization by 1.82, and a causal fourth-order 100–3,800-Hz band-pass. It is not physical replay. For nine conditions,

$$\Delta F1_k = F1_{clean} - F1_k, \quad \overline{\Delta F1} = \frac{1}{9} \sum_{k=1}^9 \Delta F1_k.$$

Negative degradation means improvement on this small subset and is not automatically robustness. Full-test stress, physical replay, and unseen evaluation are respectively NOT_RUN, NOT_AVAILABLE, and NOT_AVAILABLE.

3.7 Clean and Robustness Evaluation

Each artifact runs float32, batch one, one backend thread, and no accelerator in a sequential isolated process on the same Windows AMD64 desktop CPU. Ten warm-ups precede 50 timed repetitions. Measurements separate preprocessing, model-only, and end-to-end time and include P95, load time, throughput, and whole-process RSS. The commercial CPU name and installed RAM were not recorded. For N clips,

$$Q = \frac{N}{T_{total}}, \quad RTF = \frac{T_{processing}}{T_{audio}}.$$

RTF below one demonstrates faster-than-duration offline processing, not causal streaming or edge deployment.

The six full-test score files are joined by sample ID. Model A dominates B iff

$$f_i(A) \leq f_i(B) \forall i \quad \text{and} \quad \exists j: f_j(A) < f_j(B).$$

Table 4: Clean performance on all 2,737 canonical test recordings

Model	F1-Score	Prec.	Rec.	F1	Macro-F1	AUC	EER
MobileNetV3	.9766	.9741	.9707	.9724	.9761	.9911	.0250
ShuffleNetV2	.9850	.9795	.9854	.9824	.9847	.9929	.0146
MnasNet-A1	.9697	.9599	.9690	.9645	.9690	.9886	.0310
EfficientNet-B0	.9635	.9724	.9406	.9563	.9624	.9877	.0400
RawNet2 ext.	.4936	.4380	.6807	.5330	.4900	.5178	.4813
AASIST ext.	.5513	.4758	.5585	.5139	.5487	.5597	.4463

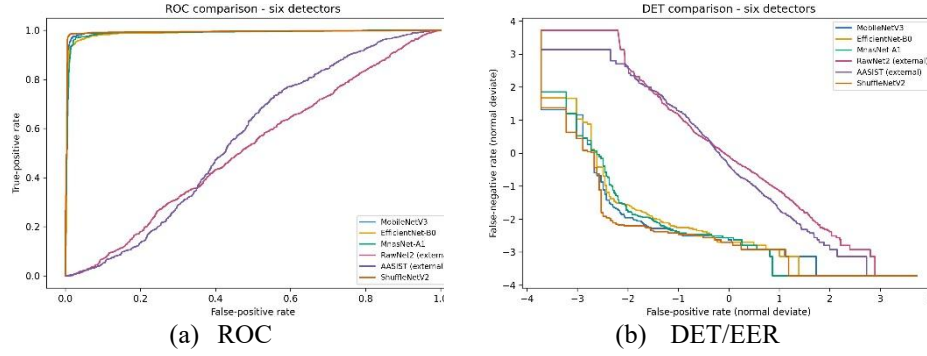


Fig. 4. Threshold-independent clean discrimination on the full test split

4 Results and Discussion

4.1 RQ1 - Clean Detection Performance

All six artifacts passed load, finite bounded-score, hash, and adapter checks. Table IV reports the full canonical test. ShuffleNet led every aggregate (F1 0.9824, macro-F1 0.9847, AUC 0.9929, EER 0.0146) with 24 false positives and 17 false negatives. MobileNet ranked second on F1 and AUC, followed by MnasNet and the available EfficientNet warm-up checkpoint. RawNet2 and AASIST were near chance under this dataset/adapter contract. This is not an architecture-only ranking because provenance, duration, representation, threshold calibration, and training differ.

Nevertheless, validation-calibrated lightweight thresholds and uncalibrated reference thresholds affect confusion metrics, while domain and preprocessing shifts affect

both thresholded and raw-score measures. The results therefore support a statement about the deployed artifacts under LAVA, not about the maximum attainable RawNet2 or AASIST performance after retraining or calibration..

4.2 RQ2 – Robustness under Practical Distortions

AWGN was the largest weakness for every high-performing lightweight system: mean noise degradation ranged from 0.3555 for ShuffleNet to 0.8053 for EfficientNet and accelerated at 5 and 0 dB.

Table 5: Diagnostic F1 degradation on the fixed 100-recording subset; lower is better

Model	Clean F1	Noise Δ	Codec Δ	Replay Δ	Mean (9)
MobileNetV3	.9756	.6722	.0095	.0620	.3099
ShuffleNetV2	.9756	.3555	.0032	.0256	.1623
MnasNet-A1	.9756	.6119	.0233	.1489	.2989
EfficientNet-B0	.9630	.8053	.0000	.0641	.3650
RawNet2 ext.	.5660	.5077	-.0007	-.0109	.2241
AASIST ext.	.5060	-.0660	-.0074	-.0682	-.0402

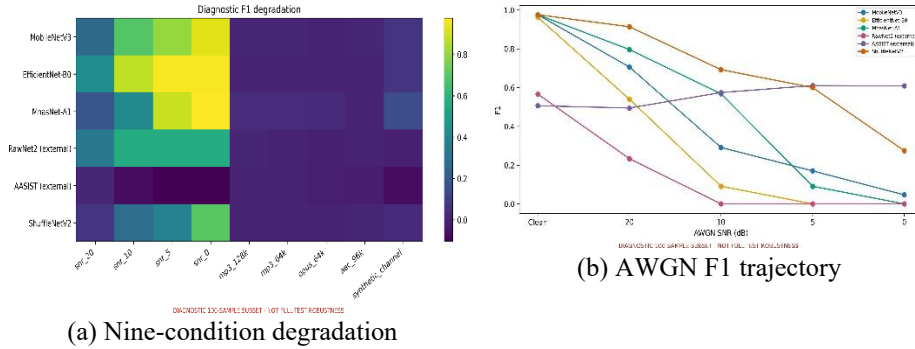


Fig. 5. Diagnostic robustness; all models reuse identical stressed recordings.

Codec changes were small for the tested settings (0.0000–0.0233 across lightweight models). Under simulated replay, ShuffleNet retained F1 0.9500 (Δ F1 0.0256), compared with degradations 0.0620/0.0641/0.1489 for MobileNet/EfficientNet/MnasNet. Negative reference deltas arise from low baselines and score redistribution; they do not establish superior robustness.

These outcomes answer RQ2 only for the executed subset and transformations. Codec stability cannot be generalized beyond four round trips, replay is not physical, and no unseen corpus exists.

4.3 Computational Efficiency

Under the common single-thread CPU protocol (Table VI), MobileNet had the lowest latency and RTF; ShuffleNet was second among TensorFlow detectors. AASIST combined the smallest artifact with the largest latency, whereas RawNet2 combined the largest parameter count with moderate latency. Parameters and serialized size therefore did not predict runtime. RSS is whole-process memory, not model-only tensor allocation.

Table 6: Measured float32 batch-one desktop-CPU efficiency (10 warm-ups, 50 runs).

Model	Params	MiB	RSS	E2E ms	P95 ms	clips/s	RTF
MobileNetV3	1.31M	5.64	554.6	43.81	52.12	22.83	.0146
ShuffleNetV2	1.87M	7.74	501.8	62.52	75.37	15.99	.0208
MnasNet-A1	3.37M	13.43	566.0	117.00	138.96	8.55	.0390
EfficientNet-B0	4.78M	18.96	651.2	175.85	196.11	5.69	.0586
RawNet2 ext.	17.62M	67.65	249.5	96.54	102.62	10.36	.0239
AASIST ext.	.30M	1.61	442.5	362.91	398.45	2.76	.0899

4.4 RQ3 – Pareto Trade-Offs

The non-dominated diagnostic set is MobileNet, ShuffleNet, and AASIST. MobileNet supplies the lowest RTF; ShuffleNet combines the same subset EER with lower degradation at a modest latency cost. AASIST continues to be non-dominated since the negative degradation due to a poor clean baseline occurs at the extreme coordinate. Frontier membership is not a scalar ranking. ShuffleNet dominates MnasNet and RawNet2; MobileNet dominates EfficientNet.

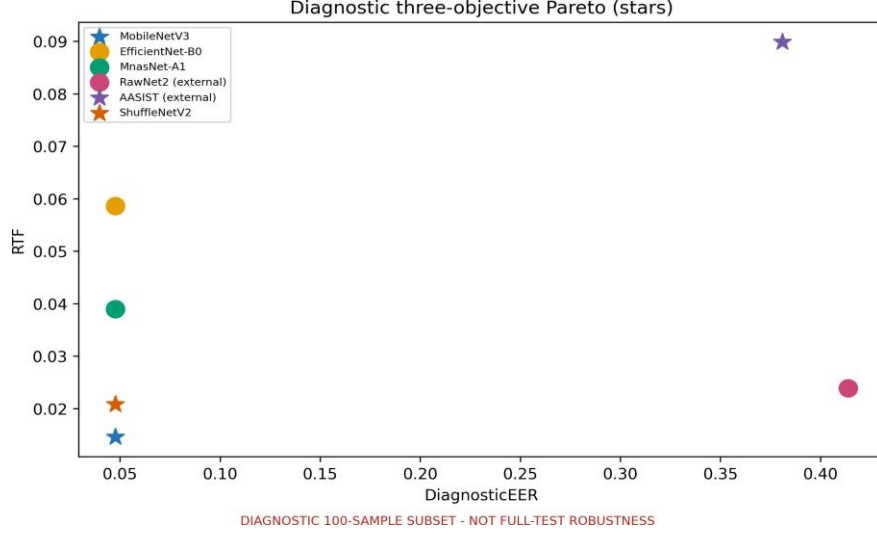


Fig. 6. EER–RTF projection; stars mark the three-objective diagnostic frontier

5 Conclusion and limitations

The study mixes ImageNet transfer (MobileNet/EfficientNet), scratch optimization (ShuffleNet/MnasNet), and external RawNet2/AASIST checkpoints, so evaluation consistency is not identical training. Missing speaker/source/generator identifiers permit only checksum-group-disjoint claims. Robustness uses 100 samples; replay is simulated; no unseen corpus, physical replay, codec-version seal, or probability calibration exists.

Runtime was measured on one desktop CPU rather than an edge target, and single checkpoints plus test-set bootstrap do not replace multi-seed training [25].

Future work should first execute full-test stresses and a metadata-rich external corpus [16], [17], then measure physical replay and target hardware before quantization or streaming studies.

LAVA provides an evidence-traceable interface for six architecturally heterogeneous voice anti-spoofing artifacts. On 2,737 clean test recordings, ShuffleNetV2-LSTM delivered the best F1/AUC/EER, while MobileNetV3Small-LSTM minimized latency and RTF. The diagnostic stresses identify low-SNR noise as the principal weakness, and multi-objective analysis retains MobileNet, ShuffleNet, and AASIST for different trade-offs. These results support reproducible deployment-oriented comparison and faster-than-duration offline inference, not unseen-domain, physical-replay, streaming, or edge-device validation.

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